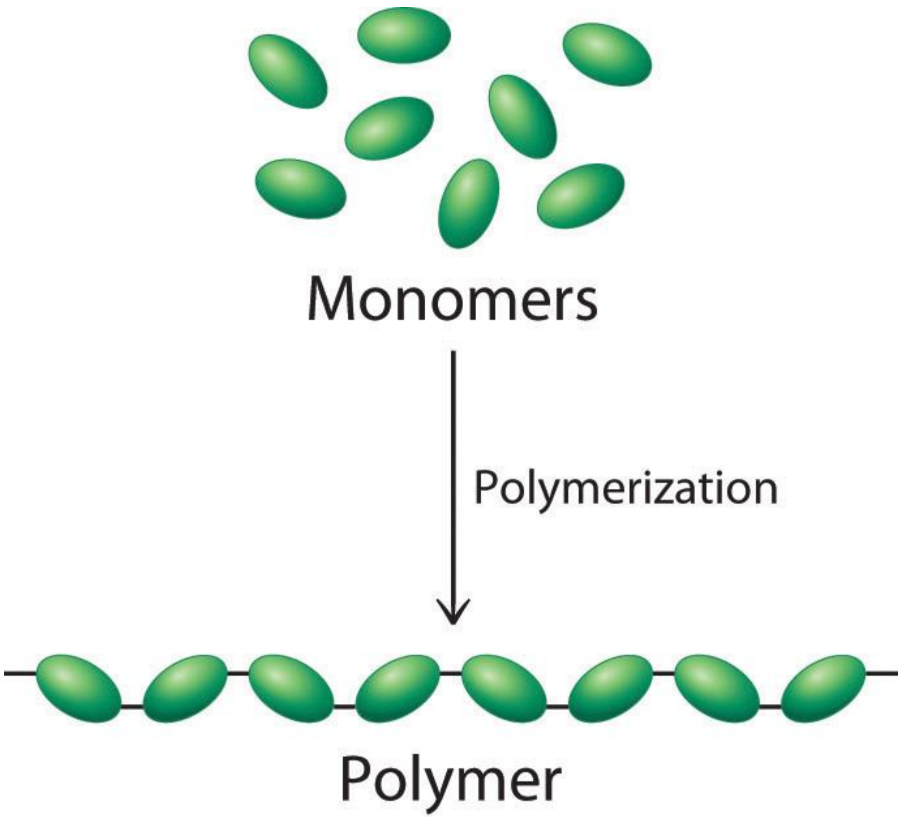
Module 4: Polymer

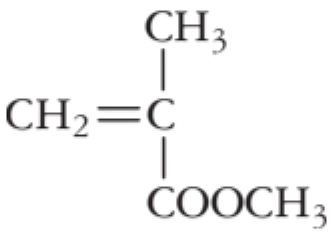
- The word polymer is derived from the classical Greek words poly meaning "many" and meres meaning "parts."
- Simply stated, a polymer is a long-chain molecule that is composed of a large number of repeating units of identical structure.

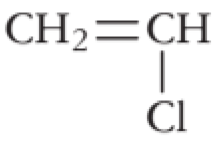


Monomers

Monomers: These are simple molecules, which combine with each other to form polymers. Monomers are also called "building blocks" of polymers.

For example, ethylene, methylmethacrylate, vinyl chloride, etc.





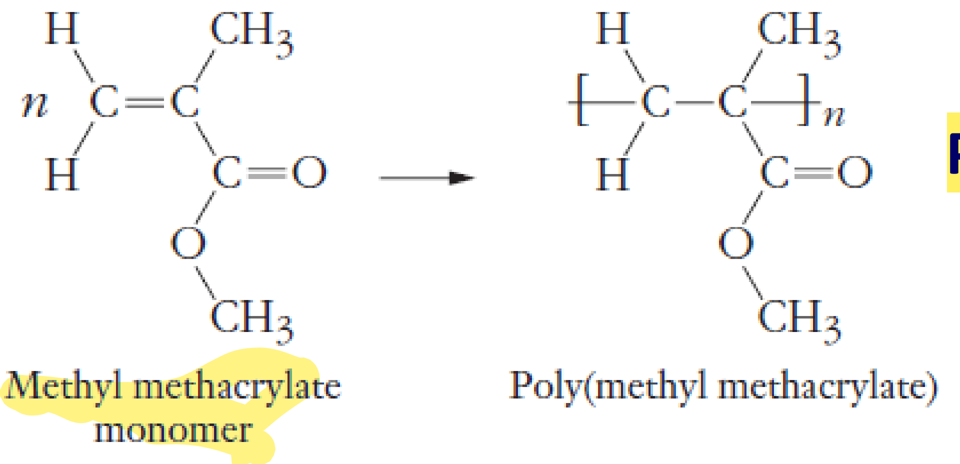
Polymer

- The monomer is the *repeat unit* of the polymer, and a typical polymer may have from hundreds to *hundreds of thousands of repeat units.*

- *Synthetic* polymers are created by chemical reactions in the laboratory; *natural* polymers (or *biopolymers*) are created by chemical reactions within organisms.

Polymerization reaction:

Monomer



Polymer

(PMMA)

Salient Features of Polymerization Reaction:

1. Generally a polymerization requires ***initiator***, in this case a peroxide, is added to trigger the process.
In some cases *Lewis acid or Bronsted* acid is used to *initiate and speed-up* the polymerization process.

2. In simple *addition polymerization*, *n-monomers are added* together to form a polymer chain with ***n-repeat units, n is very large 100s or 1000s***

Salient Features of Polymerization Reaction *(continued)*

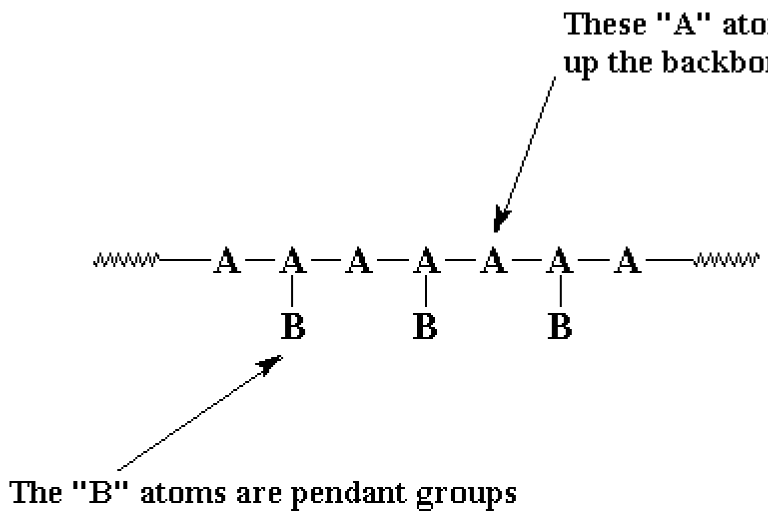
3. Polymerization normally *occurs in an uncontrolled manner*. Therefore, the molecular weight of the polymers will not be uniform. Hence, the ***molecular weight of a polymer is an average quantity***.

4. The *molecular weights of polymers are much larger than the small molecules* usually encountered in organic chemistry.

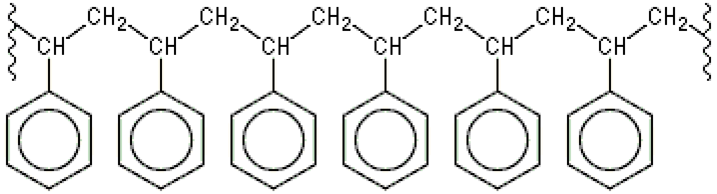
5. The high molecular weights of the polymers ensure their mechanical properties*, toughness, rigidity, plasticity*, and their *thermal softening* properties.

Structural Features of Polymer

Each polymer chain will have a backbone and may have pedant groups (side chain or branching atoms).



Example-Polystyrene



polystyrene

Backbone consists of alternating CH_2 and CH -CH_2-CH-CH_2-CH- repeat unit *with all carbon atoms* in the backbone.
Phenyl group is the pendant group

Molecular weight

Molecular weight

- Due to the uncontrolled polymerization process, a typical synthetic polymer sample contains chains with a wide statistical distribution of chain lengths.

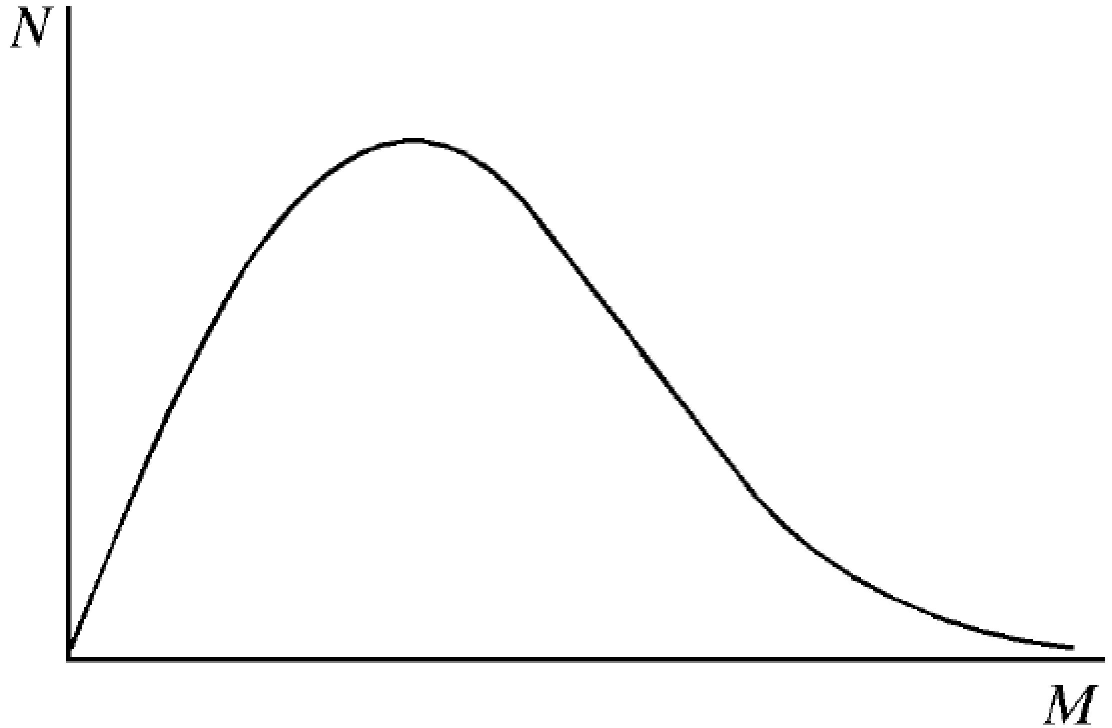
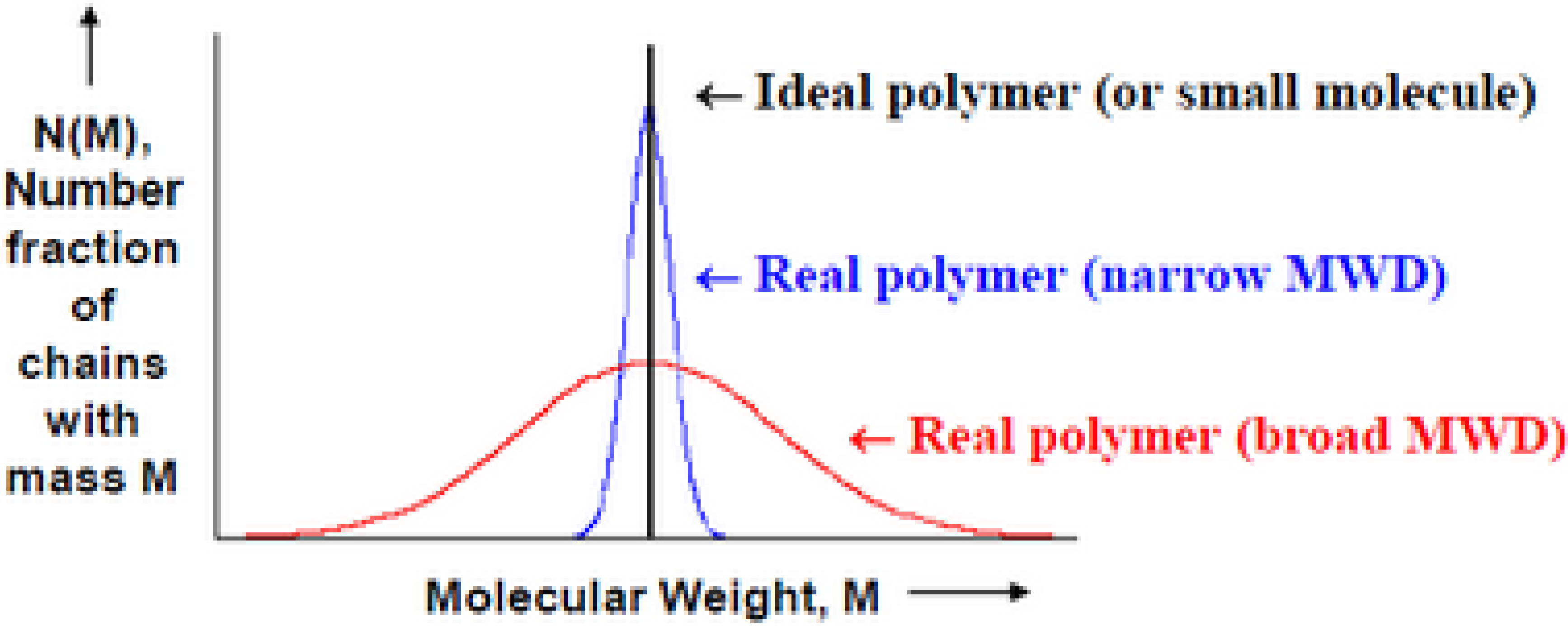


Figure: A representation of a continuous distribution of molecular weights shown as a plot of the number of moles of chains, N, having molecular weight M,against M.

Molecular weight

- The polymerization of some monomers can result in molecular-weight distributions that are extremely broad.
- In some polymerizations, polymers with very narrow molecular-weight distributions can be obtained.
- polymer properties, such as melt viscosity and elasticity, are dependent on molecular weight and molecular-weight distribution.
- Therefore, it is useful to define molecular-weight averages associated with a given molecular-weight distribution


N(M),
Number
fraction
of
chains
with
mass M
← Ideal polymer (or small molecule)
← Real polymer (narrow MWD)
← Real polymer (broad MWD)
Molecular Weight, M

Molecular weights of polymers: An average quantity

- Number-average molecular weight

$$\bar{M}_n = \frac{\sum_{i=1}^{N} N_i M_i}{\sum_{i=1}^{N} N_i} = \frac{\sum_{i=1}^{N} W_i}{\sum_{i=1}^{N} (W_i / M_i)}$$

N_i = Number of moles of molecules having molecular weight of M_i

$\sum_{i=1}^{N} N_i$ = *Total number of moles of entire sample*

- Weight-average molecular weight

$$\bar{M}_w = \frac{\sum_{i=1}^{N} N_i M_i^2}{\sum_{i=1}^{N} N_i M_i} = \frac{\sum_{i=1}^{N} W_i M_i}{\sum_{i=1}^{N} W_i}$$

$$N_i = W_i/M_i$$

Calculation of Number Average and Weight Average Molecular Weight

S.No	Number of Molecules or moles	Molecular Weight gm/mol
1	200	5000
2	300	10000
3	400	20000
4	100	40000

$$\text{Number Average Molecular Weight } \overline{M}_n = \frac{\sum_{i=1}^{N} N_i M_i}{\sum_{i=1}^{N} N_i}$$

$= 16000 \text{ gm mol}^{-1}$

Calculate the weight-average molar mass

S.No	Number of Molecules	Molecular Weight gm/mol
1	200	5000
2	300	10000
3	400	20000
4	100	40000

$$\text{Weight Average Molecular Weight } \overline{M}_w = \frac{\sum_{i=1}^{N} N_i M_i^2}{\sum_{i=1}^{N} N_i M_i}$$

$= 22200 \text{ gm mol}^{-1}$

Mw > Mn

Polydispersity Index (PDI)

Ratio of M_w and M_n is known as Polydispersity Index (PDI)

$$\text{Polydispesity index} = \frac{M_w}{M_n}$$

Polydispersity index for the previous sample data.

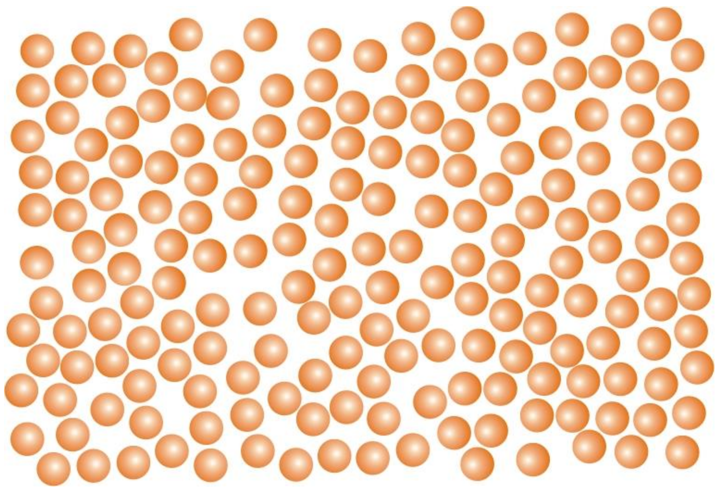
22200/16000 = 1.39

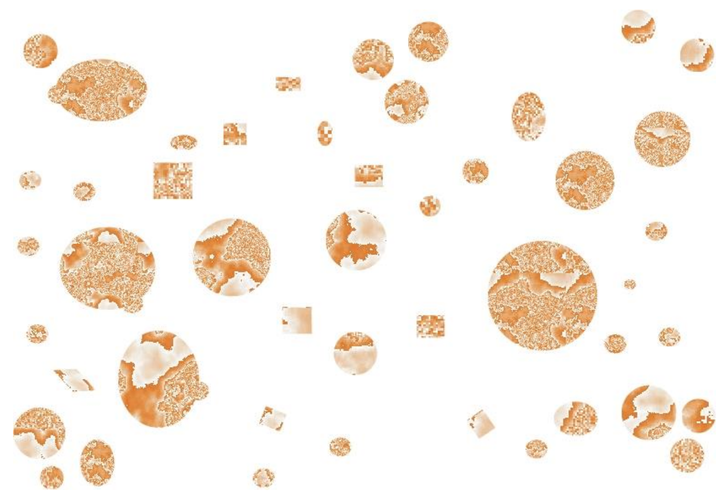
Higher the PDI, more spread is the molecular weight.

Many of the synthetic polymers have PDI > 1 (Polydisperse, Mw > Mn)
Biopolymers DNA, RNA, Proteins have PDI = 1 (Monodisperse, Mn = Mw)

Polydispersity Index (PDI)

- Which one of the following has high polydispersity index?





Which one of the following polymers does not have monodispesity (PDI ~ 1) ?

A) Poly (methyl methacylate)

B) DNA

C) RNA

D) Hemoglobin

Ans: Poly (methyl methacrylate)

Problem 1

A polydisperse sample of polystyrene is prepared by mixing three *monodisperse* samples in the following proportions:

1 g	10,000 molecular weight
2 g	50,000 molecular weight
2 g	100,000 molecular weight

Using this information, calculate the number-average molecular weight, weight-average molecular weight, and PDI of the mixture.

Problem 1

A polydisperse sample of polystyrene is prepared by mixing three *monodisperse* samples in the following proportions:

1 g	10,000 molecular weight
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2 g	100,000 molecular weight

Using this information, calculate the number-average molecular weight, weight-average molecular weight, and PDI of the mixture.

$$\bar{M}_{\mathrm{n}} = \frac{\sum_{i=1}^{3} N_i M_i}{\sum_{i=1}^{3} N_i} = \frac{\sum_{i=1}^{3} W_i}{\sum_{i=1}^{3} \left(W_i / M_i\right)} = \frac{1+2+2}{\frac{1}{10,000} + \frac{2}{50,000} + \frac{2}{100,000}} = 31,250$$

$$\bar{M}_{\mathrm{w}} = \frac{\sum_{i=1}^{3} N_i M_i^2}{\sum_{i=1}^{3} N_i M_i} = \frac{\sum_{i=1}^{3} W_i M_i}{\sum_{i=1}^{3} W_i} = \frac{10,000 + 2\left(50,000\right) + 2\left(100,000\right)}{5} = 62,000$$

$$\mathrm{PDI} = \frac{\bar{M}_{\mathrm{w}}}{\bar{M}_{\mathrm{n}}} = \frac{62,000}{31,250} = 1.98$$

Problem 2

A polymer has the following composition: 100 molecules of molecular mass 1000 g/mol, 200 molecules of molecular mass 2000 g/mol and 500 molecules of molecular mass 5000 g/mol. Calculate the number and weight average molecular weight and the polydispersity index.

Problem 2

Given that $M_1 = 1000$ g/mol, $N_1 = 100$; $M_2 = 2000$ g/mol, $N_2 = 200$; $M_3 = 5000$ g/mol, $N_3 = 500$. The number average molecular weight in given by:

$$\overline{M_n} = \frac{\sum N_i M_i}{\sum N_i} = \frac{100\times 1000 + 200\times 2000 + 500 + 5000}{100 + 200 + 500} = \frac{1\times 10^5 + 4\times 10^5 + 25\times 10^5}{800}$$

$$= 3.75\times 10^3 \text{g/mol}$$

The weight average molecular weight is given by:

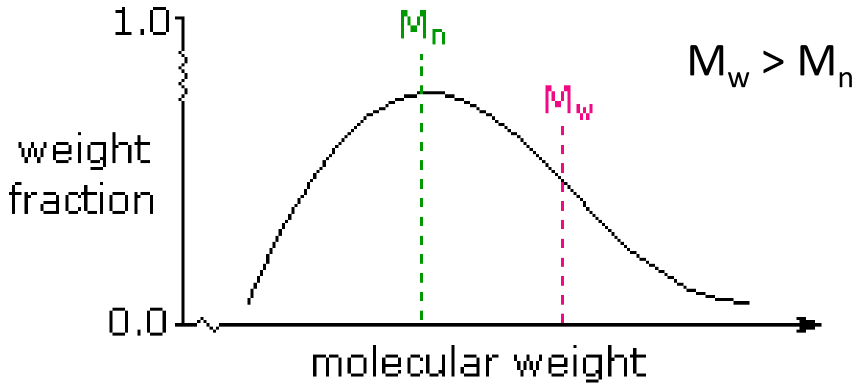
$$\overline{M_w} = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{100\times (1000)^2 + 200\times (2000)^2 + 500\times (5000)^2}{30\times 10^5}$$

$$= \frac{1\times 10^8 + 8\times 10^8 + 125\times 10^8}{30\times 10^5} = \frac{134\times 10^8}{30\times 10^5} = 4.46\times 10^3 \text{ g/mol}$$

The polydispersity index (PDI) is

$$\frac{\overline{M_w}}{\overline{M_n}} = \frac{4.46\times 10^3}{3.75\times 10^3} = 1.19$$

M_w is always greater than M_n



M_n and M_w in a typical sample of polydispersed macromolecules

Since *larger molecules in a sample weigh more than smaller molecules*, the weight average M_w is necessarily skewed to higher values, and is always greater than M_n.

As the weight dispersion of molecules in a sample narrows, M_w approaches M_n, and in the unlikely case that all the polymer molecules have identical weights (a pure mono-disperse sample), the ratio *Mw/Mn becomes unity*.

Degree of polymerization (DP)

- Degree of polymerization (DP = n) is defined as the ratio between the molar mass of the polymer and molar mass of the repeat unit.

$$\text{Degree of polymerization} = \frac{\text{molar mass of polymer}}{\text{molar mass of monomer}}$$

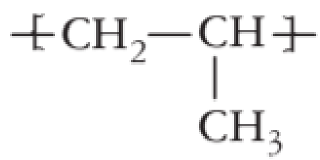
Table 12.8 Molar Masses of Some Common Polymers

Name	$\mathcal{M}_{\text{polymer}}$ (g/mol)	n	Uses
Acrylates	2×10^5	2×10^3	Rugs, carpets
Polyamide (nylons)	1.5×10^4	1.2×10^2	Tires, fishing line
Polycarbonate	1×10^5	4×10^2	Compact discs
Polyethylene	3×10^5	1×10^4	Grocery bags
Polyethylene (ultrahigh molecular weight)	5×10^6	2×10^5	Hip joints
Poly(ethylene terephthalate)	2×10^4	1×10^2	Soda bottles
Polystyrene	3×10^5	3×10^3	Packing, coffee cups
Poly(vinyl chloride)	1×10^5	1.5×10^3	Plumbing

Problem

If average degree of polymerization of polypropylene is 2×10^4, calculate the average molecular weight of the polymer.

Solution

The repeat unit in polypropylene is: 

Molecular weight of repeat unit is = 42 g/mol. We know that

Molecular weight of polymer = Molecular weight of repeat unit × Degree of polymerization

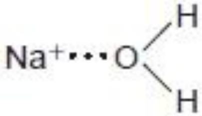
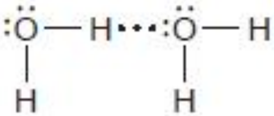
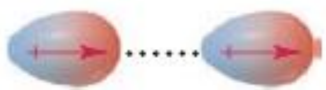
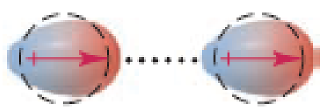
Therefore,

$$\text{Molecular weight of polymer} = 42 \times 2 \times 10^4 = 84 \times 10^4 = 8.4 \times 10^5 \text{ g/mol}$$

The polymer molecular weight obtained in this way is the number average molecular weight (M_n).

Structure property relations
-Intermolecular force
-Crystalline Vs Amorphous
-Glass Transition Properties
of Polymers

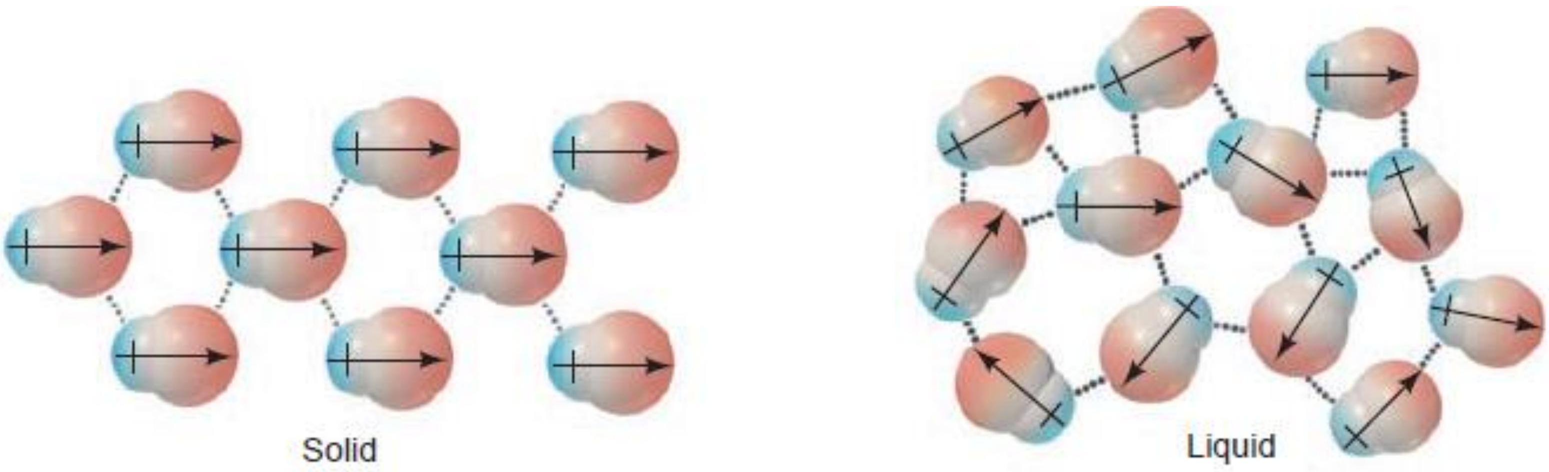
Intermolecular forces

Force	Model	Basis of Attraction	Energy (kJ/mol)	Example
Nonbonding (Intermolecular)				
Ion-dipole		Ion charge–dipole charge	40–600	
H bond	$\overset{\delta-}{\text{—A}}\text{—}\overset{\delta+}{\text{H}}\cdots\cdots\cdot\!:\!\overset{\delta-}{\text{B}}\text{—}$	Polar bond to H–dipole charge (high EN of N, O, F)	10–40	
Dipole-dipole		Dipole charges	5–25	I—Cl···I—Cl
Dispersion (London)		Polarizable e^- clouds	0.05–40	F—F···F—F

Intermolecular forces

- **Dipole-Dipole Forces**

When polar molecules lie near one another, as in liquids and solids, their partial charges act as tiny electric fields that orient them and give rise to **dipole-dipole forces**.

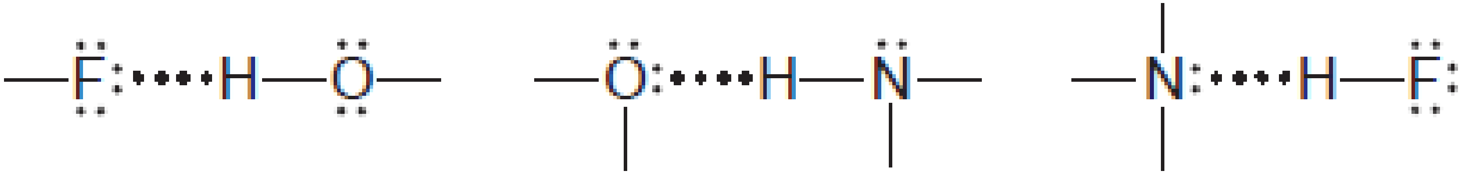


Intermolecular forces

- **The Hydrogen Bond**

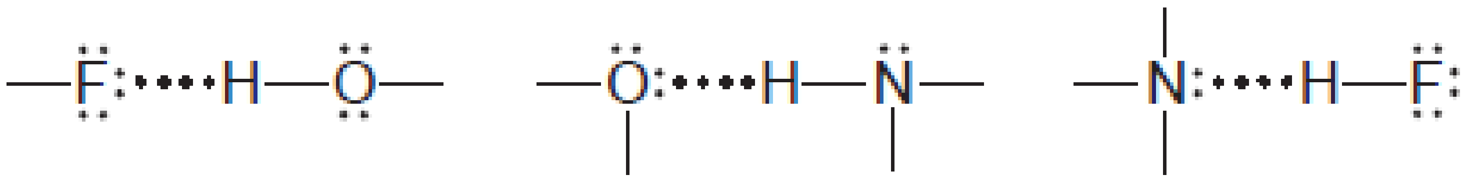
Hydrogen bond forms between molecules that have an H atom bonded to a small, highly electronegative atom with lone electron pairs.

The most important atoms that fit this description are N, O, and F.



Intermolecular forces

- **The Hydrogen Bond**



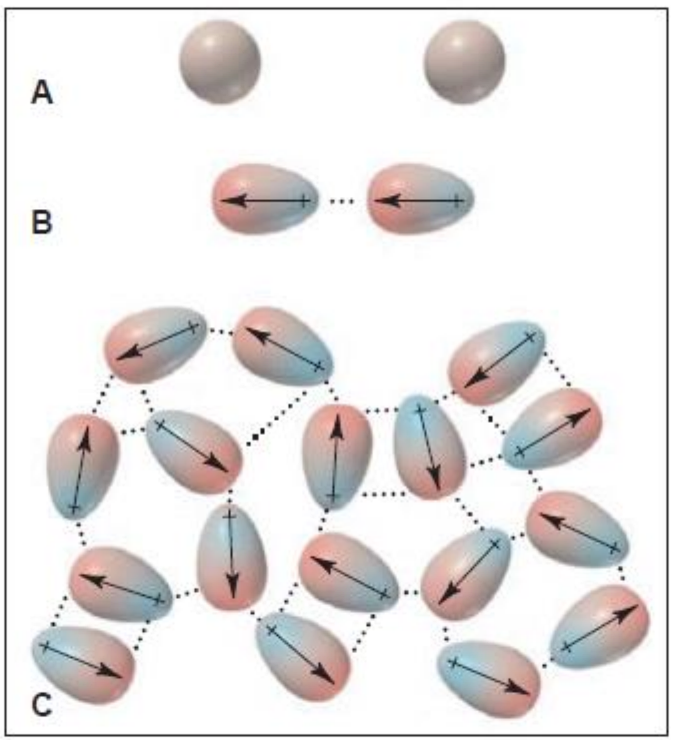
- The H—N, H—O, and H—F bonds are very polar, so electron density is withdrawn from H. As a result, the partially positive H of one molecule is attracted to the partially negative lone pair on the N, O, or F of another molecule

- Hydrogen bonding is a strong type of dipole-dipole force

Intermolecular forces

Dispersion forces

Dispersion forces are caused by momentary oscillations of electron charge in atoms and, therefore, are present between all particles (atoms, ions, and molecules).



Separated Ar atoms are nonpolar

An instantaneous dipole in one atom induces a dipole in its neighbour. These partial charges attract the atoms together.

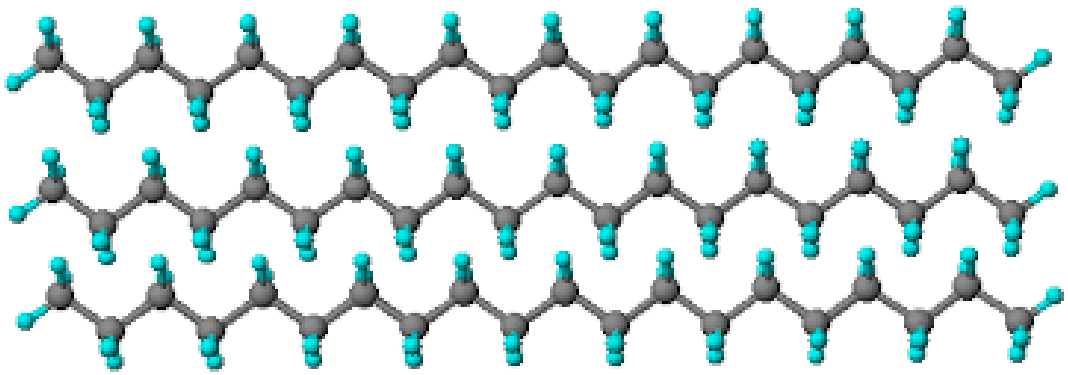
This process takes place among atoms throughout the sample.

nonpolar molecules show dispersion forces.

Intermolecular force

Weak force of attraction between the molecules

Inter polymer chain or intermolecular forces



Polyethylene – Long chain hydrocarbon

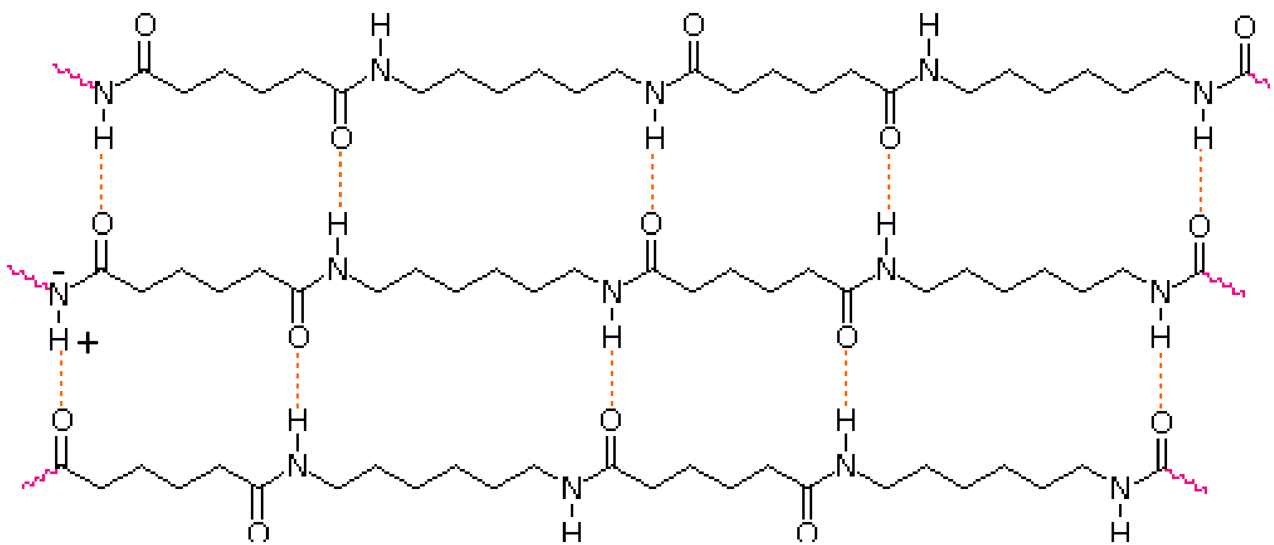
Instantaneous dipole –induced dipole

Or dispersion interaction

Polyamides

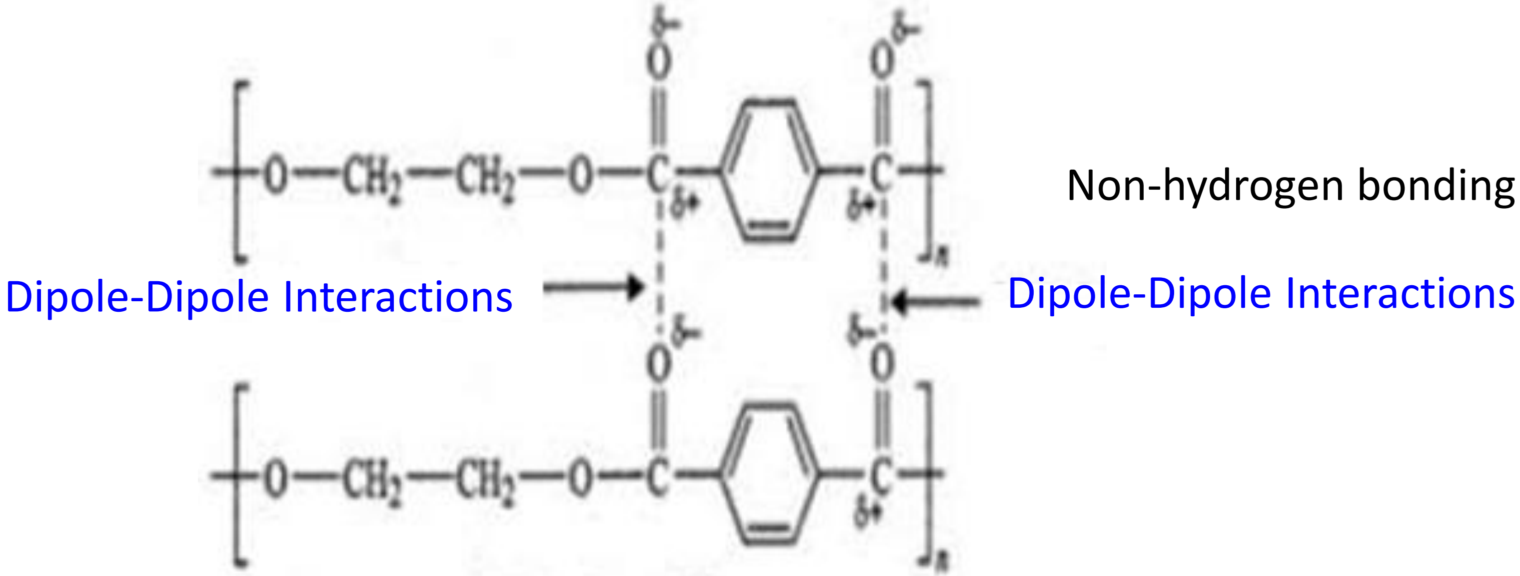
Hydrogen bonding

Permanent dipole



Polyester

Dipole-Dipole Interactions



The extent of slipping away of the chains from one another depends on the strength of the interchain force.

Strength of intermolecular force: Hydrogen bond > Dipole-Dipole > Dispersion force

"Tendency of chains to slip away is in reverse order"

Hydrogen bonded chains have less tendency to slip away than that of dipole-dipole bonded systems.

"Therefore hydrogen bonded systems are tough fibers with large stiffness and elastic modulus". (Example cellulose)

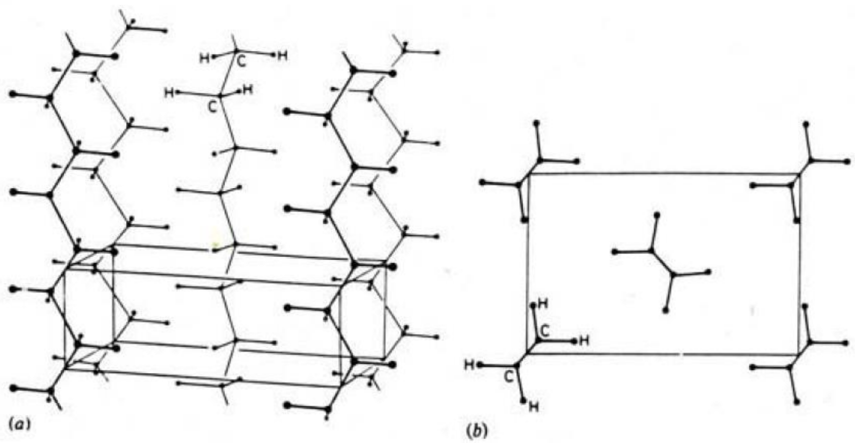
Chemical structure- Crystalline Vs Amorphous

Crystallinity:

The requirement for a polymer to crystallize is regularity of molecular structure with greater symmetry, such that polymer chains can pack up closely in an ordered structure.

The degree of crystallinity of the polymer depends on its structure.

Examples linear polyethylene and isotactic polypropylene



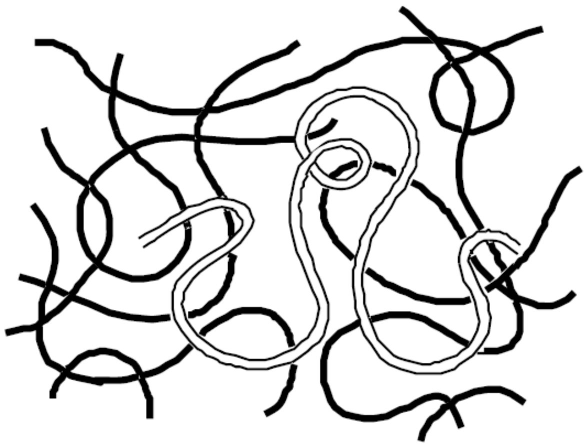
Crystalline polyethylene: orthorhombic

In crystalline polymers, the atomic arrangements follows a repeating pattern in long range order.

Chemical structure -Crystalline Vs Amorphous

Amorphos polymers

In many cases, individual chains are randomly coiled and intertwined with no molecular order or structure. Such a physical state is termed amorphous.



Example: atactic polystyrene

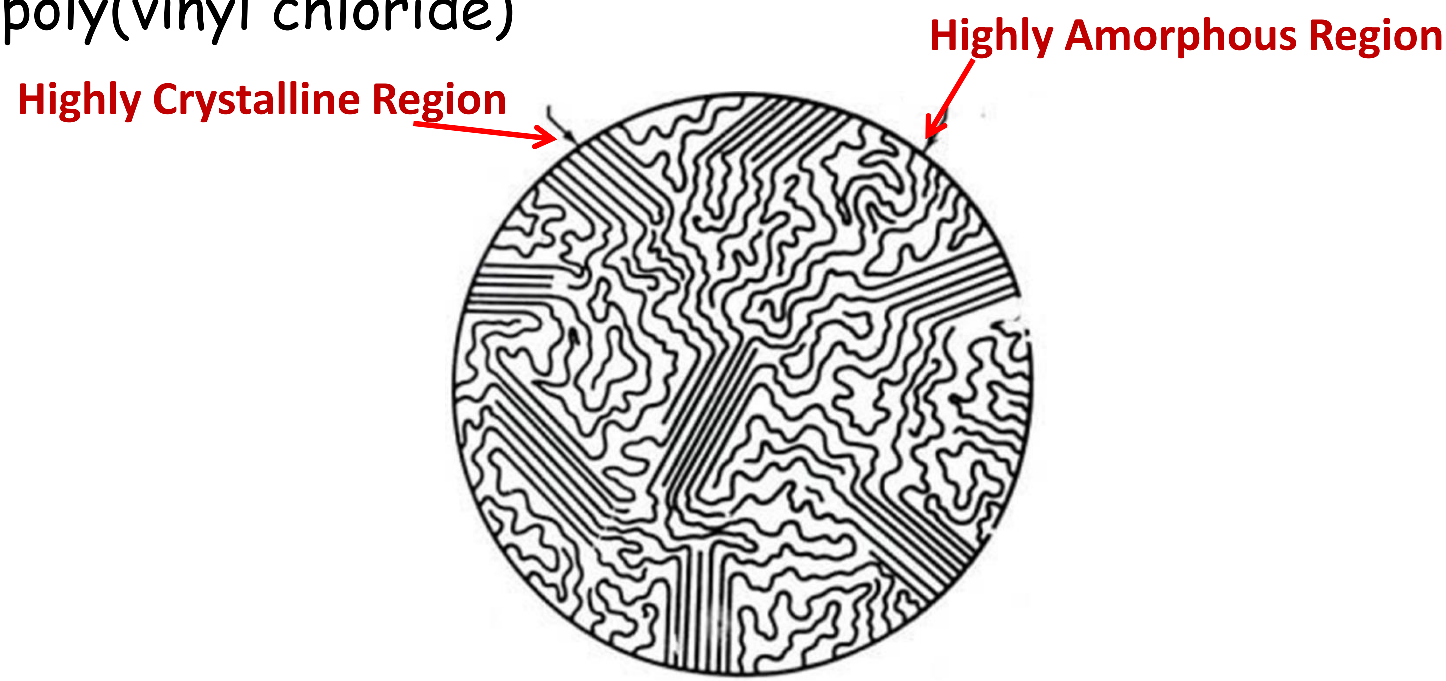
Chemical structure- Crystalline Vs Amorphous

Semicrystallinity:

- Although single crystals of some polymers such as polyethylene can be grown under laboratory conditions, no bulk polymer is completely crystalline.

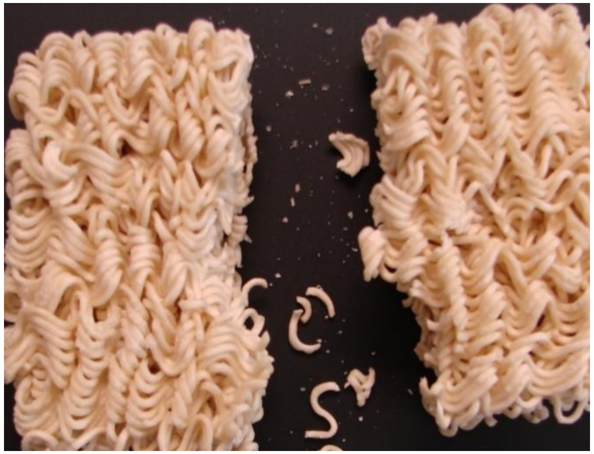
- In the case of semicrystalline polymers, regular crystalline units are linked by unoriented, random-conformation chains that constitute amorphous regions.

Examples poly(vinyl chloride)



Concept Examples : Morphology

Think of Semi-crystalline materials like maggi noodles. When in their solid state, they have a compact ordered arrangement



Amorphous materials are like cooked noodles in that there is a random arrangement

Polymers Classification – On Thermal Processing

Based on thermal processing behaviour polymers can be classified into two categories, thermoplastic or thermosets.

Thermoplastics

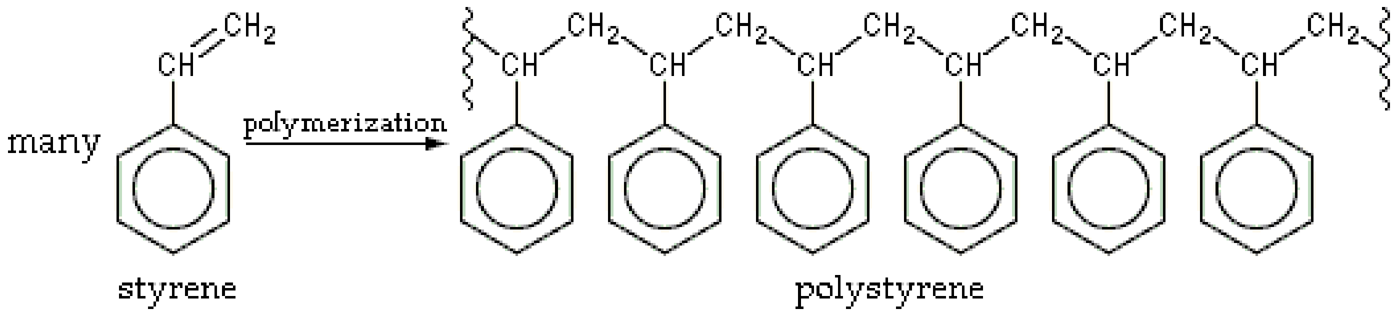
Polymers that can be heat-softened in order to process into a desired form (or shape) are called *thermoplastics*.

Examples: Polystyrene, polyolefins (e.g., polyethylene and polypropylene) and poly(vinyl chloride).

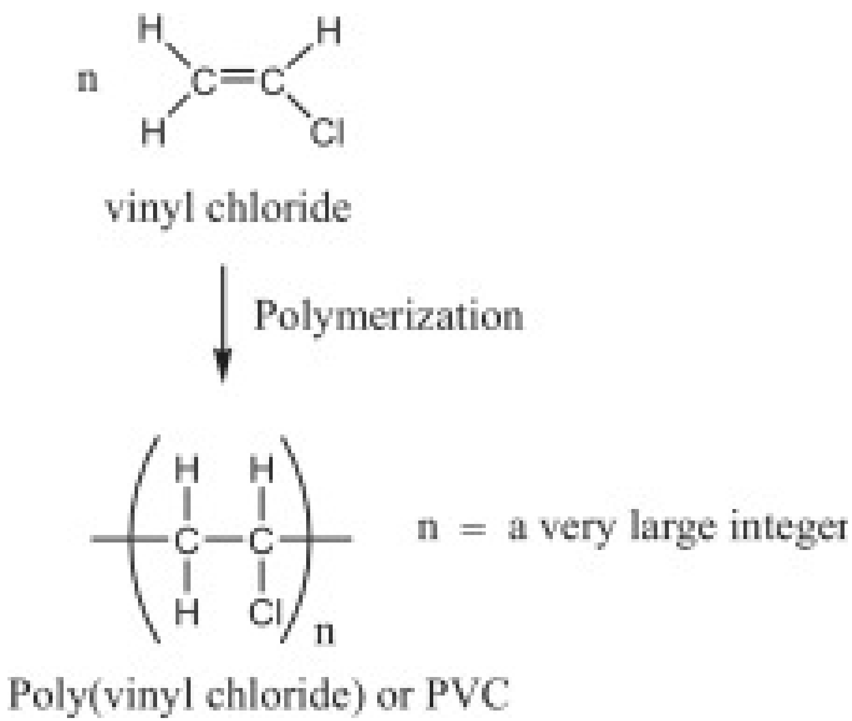
Polymers-Classification

Thermoplastics

Polystyrene



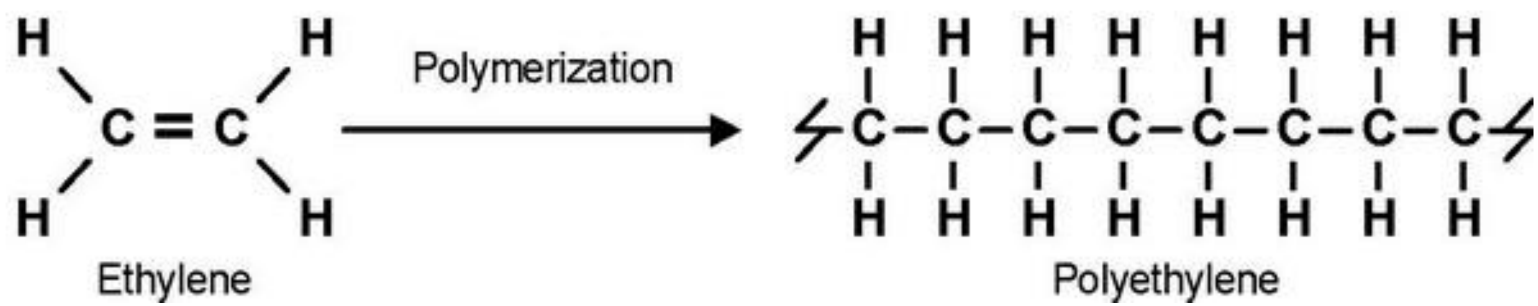
Polyvinyl chloride



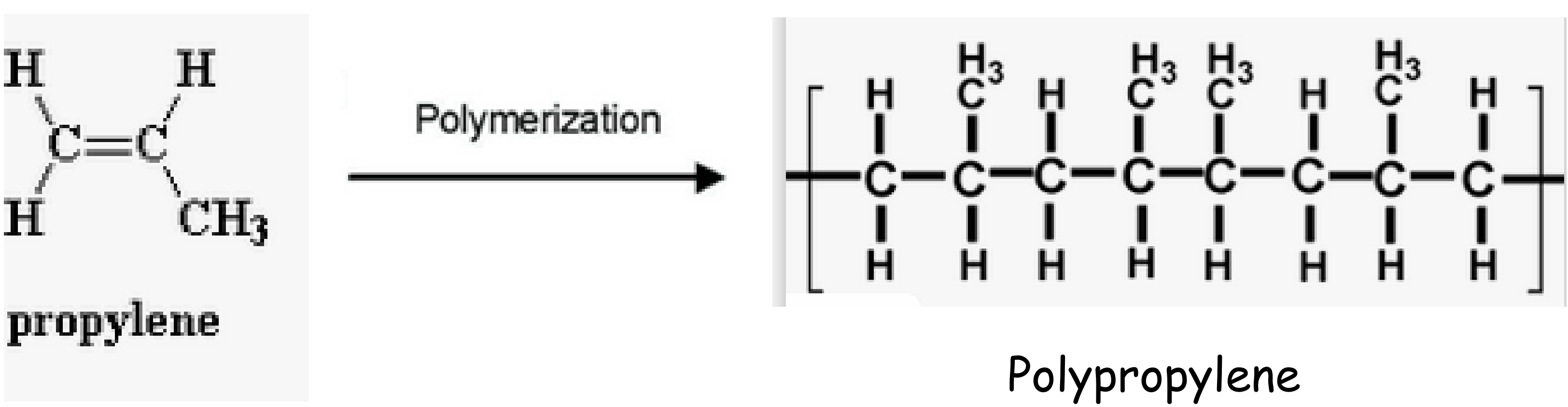
Polymers-Classification

Thermoplastics

Polyethylene



Polypropylene



Polypropylene

Polymers-Classification

Thermoset (Thermosetting)

Some polymers undergo chemical changes and cross-linking on heating and become permanently hard, rigid and infusible on cooling. These are called thermosetting polymers.

Once formed, these cross-linked networks resist heat softening, mechanical deformation, and solvent attack, but cannot be thermally processed.

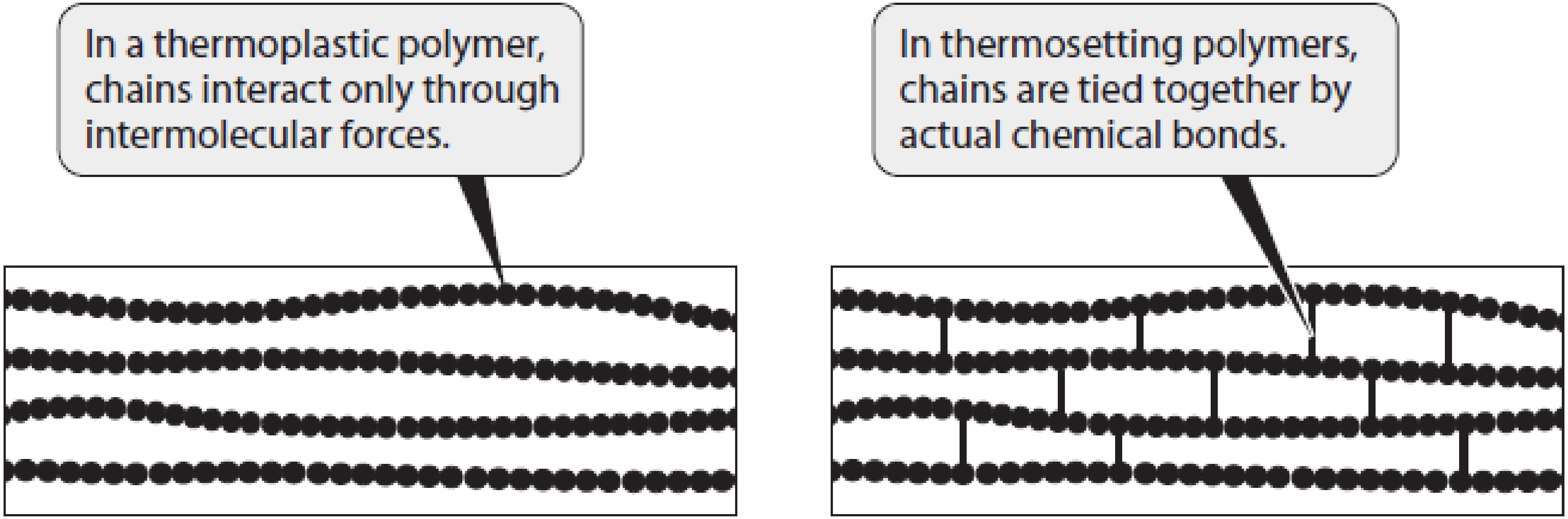
Polymers-Classification

Thermoset (Thermosetting)-Applications

Such properties make thermosets suitable materials for composites, coatings, and adhesive applications.

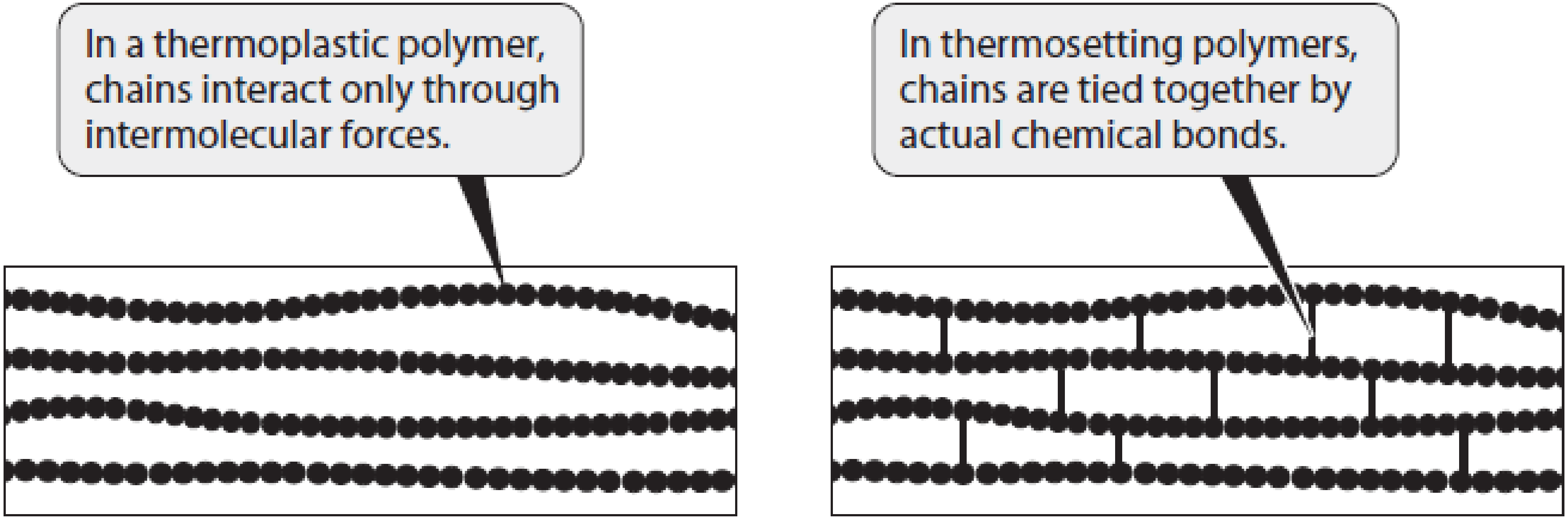
Principal examples of thermosets include epoxy, phenol–formaldehyde resins, and unsaturated polyesters that are used in the manufacture of glass-reinforced composites such as Fiberglass

Additional information on thermoplastic and thermoset



The initial heating and setting of a thermosetting polymer produces a number of links between sites on the carbon backbone of different molecular chains. These links are referred to as crosslinks because they cross between and link individual molecular strands of the polymer.

Additional information on thermoplastic and thermoset

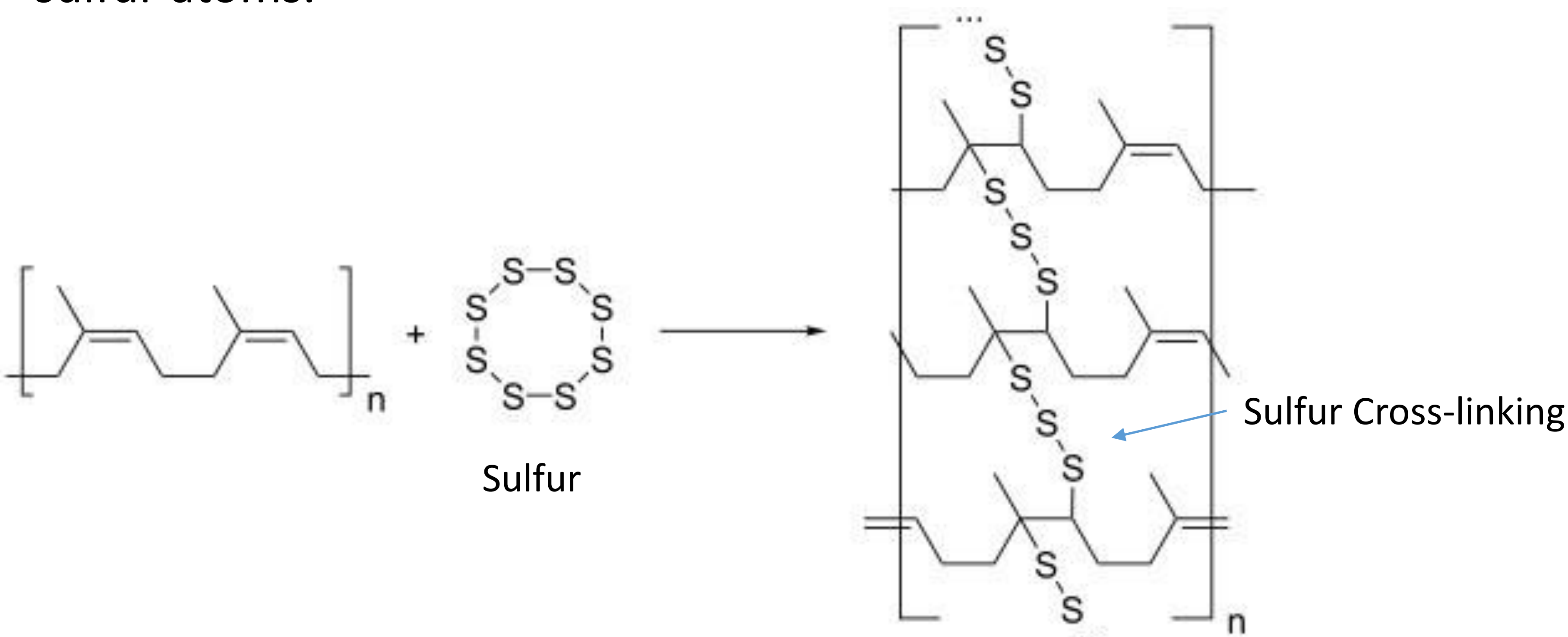


Chemically, these cross-links are additional covalent bonds that join the polymer chains to one another. Like most covalent bonds, they are strong enough that they do not readily fail upon heating. So the cross-linked polymer keeps its shape.

Example: vulcanization

Methods to Improve the Elasticity -Vulcanization

Vulcanization of rubber is a process of improvement of the rubber elasticity and strength by heating it in the presence of sulfur, which results in three-dimensional cross-linking of the chain rubber molecules (polyisoprene) bonded to each other by sulfur atoms.

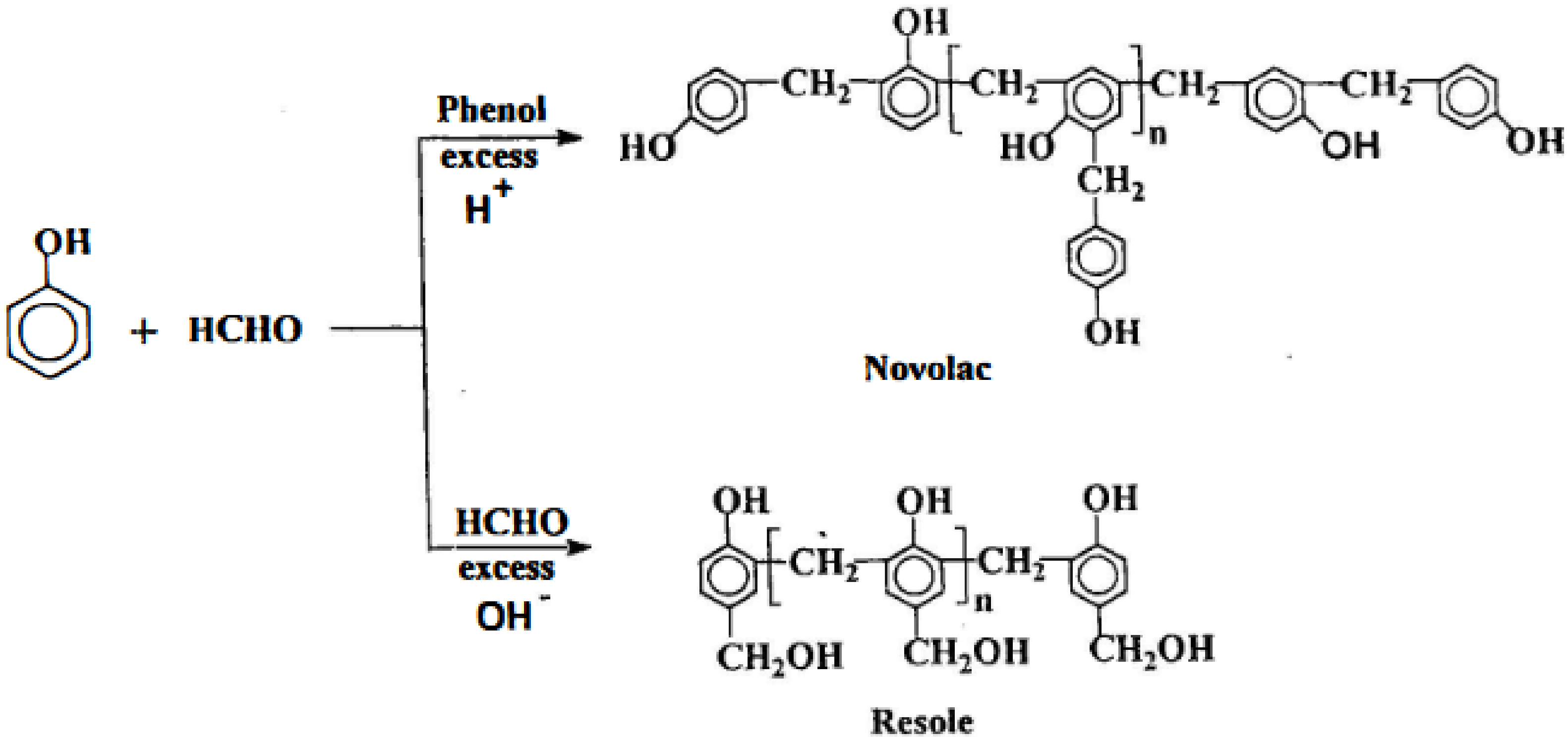


Due to cross-linking, volcanized rubbers, the cross-linked rubber can be molded into well defined shape than the unfinished natural rubber.

Phenolic resins: Bakelite

- Bakelite was the first plastic made from synthetic components.
- It is a thermosetting phenol formaldehyde resin, formed from a condensation reaction of phenol with formaldehyde.

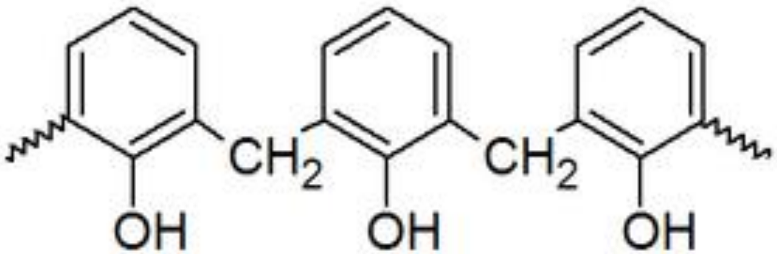
Synthesis of Bakelite



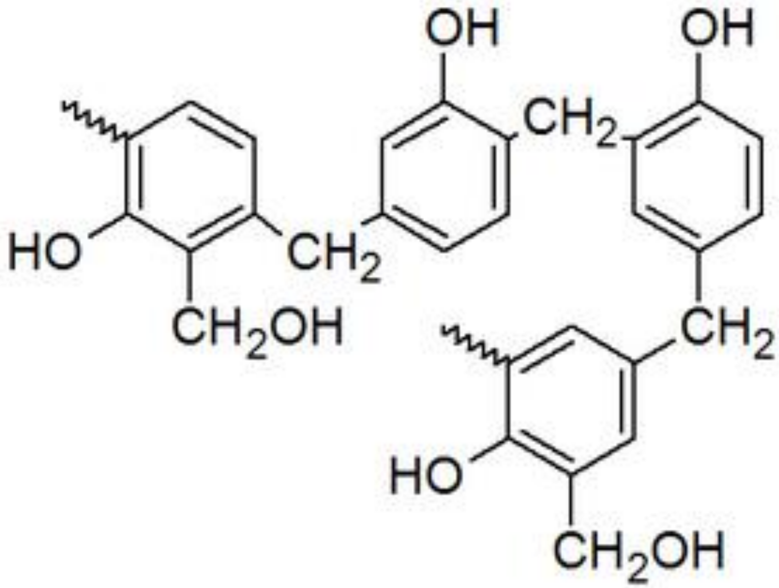
A-stage $\xrightarrow{\text{heating}}$ B-stage $\xrightarrow{\text{heating}}$ C-stage

first stage of reaction, no cross-linking	second stage of reaction, partially cross-linking	third stage of reaction, more cross-linking

Synthesis of Bakelite



Novolac

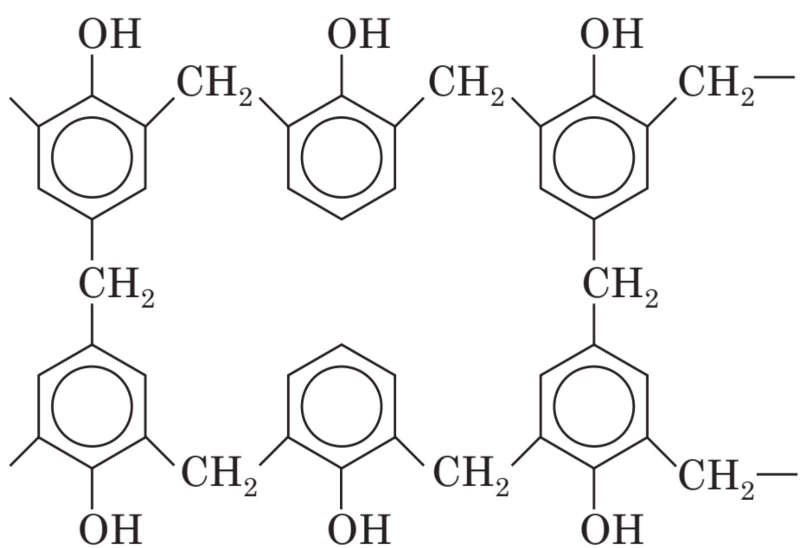


Resole

Synthesis of Bakelite

Bakelite is cross-linked polymer

$$\text{Novolac} \xrightarrow{\text{cross-linking}} \text{Bakelite}$$



bakelite

C-Stage Resin

Bakelite: Properties and Uses

Properties. These phenolic resins are rigid, hard, water resistant. They are resistant to non-oxidising acids, organic solvents but are susceptible to alkalies. Solubilities and melting point of the resin gradually change with rise of molecular weight. These resins possess electrical insulating properties.

Uses. It can be widely used as metal substitute where high tensile strength is not necessary. As an inert material it can substitute for glass. It can be used for making insulator parts like switches, plugs, heater handles. It can be moulded to cabinets for TVs and radio and telephone parts. It is used as adhesive also used in paints and varnishes, as cation exchanger resin for water softening, in paper industry as propeller shafts.




Polymers Classification
-On mechanical Properties

Elastomers, Plastics, and Fibers

Elastomers

Elastomers: The polymers which undergo high degree of elongation when pulled apart, and return to their original length on release are called elastomers.

These are mainly coiled and long-chain polymers with no intermolecular forces except weak van der Waals forces (dispersion forces).

For example, natural rubber, Buna-S, butyl rubber, silicone rubber, etc.

Polymers must have little internal hindrance to the random motion of their monomer subunits (in other words, they must not be glassy), and they must not spontaneously crystallize (at least at normal temperatures).

On release from being extended, they must be able to return spontaneously to a disordered state by random motions of their repeating units as a result of rotations around the carbon-carbon bond.

Linear vs Cross linked elastomers

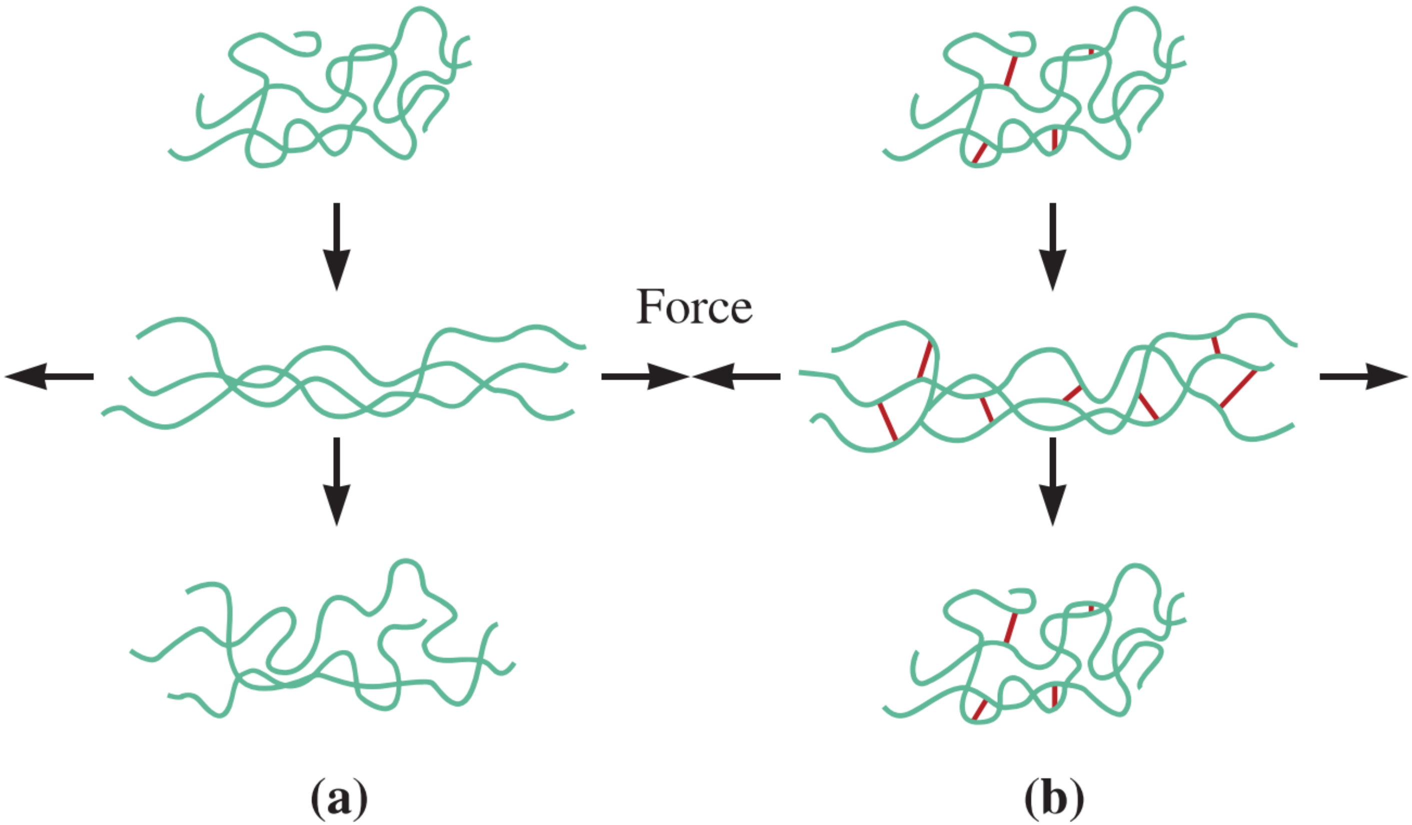


Figure 16-21 (a) When the elastomer contains no cross-links, the application of a force causes both elastic and plastic deformation; after the load is removed, the elastomer is permanently deformed. (b) When cross-linking occurs, the elastomer still may undergo large elastic deformation; however, when the load is removed, the elastomer returns to its original shape.

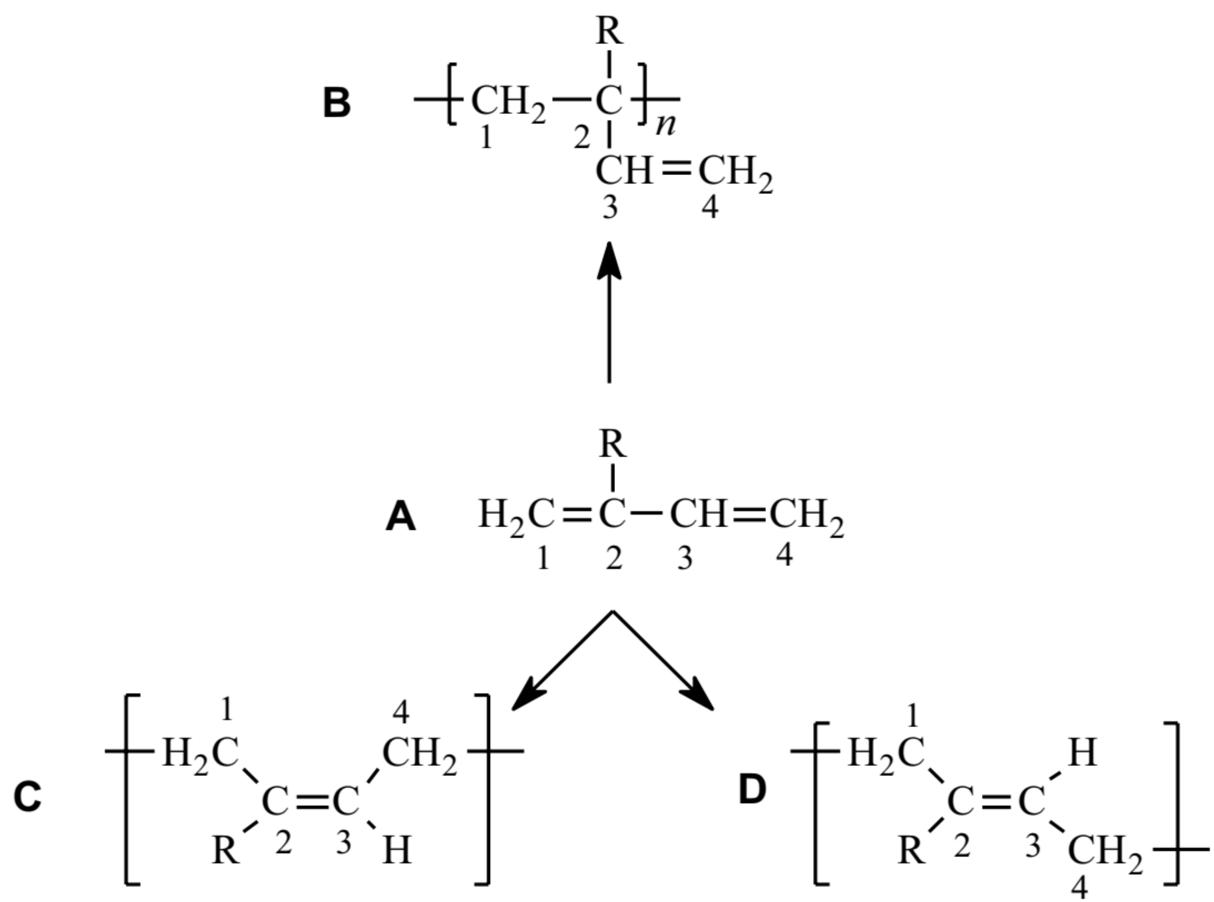
"Small degree of cross-links in elastomers acts as a memory to return to its original shape and size upon release of stress "

Elastomers [Rubbers]

A number of natural and synthetic polymers called elastomers display a large amount (>200%) of elastic deformation when a force is applied. Rubber bands, automobile tires, O-rings, hoses, and insulation for electrical wires are common uses for these materials. Crude natural rubber, which is an elastomer, can erase pencil marks; hence, elastomers got the name rubber.

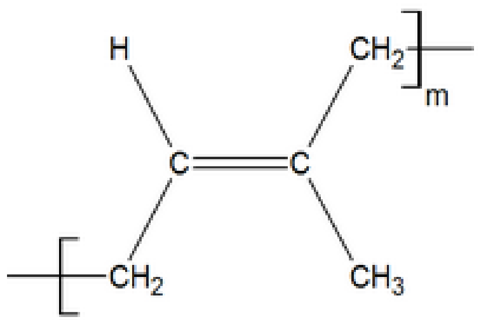
- Elastomers can be broadly classified as belonging to one of three groups:
  - Diene elastomers,
  - Non-diene elastomers, and
  - Thermoplastic elastomers.

Diene Elastomers

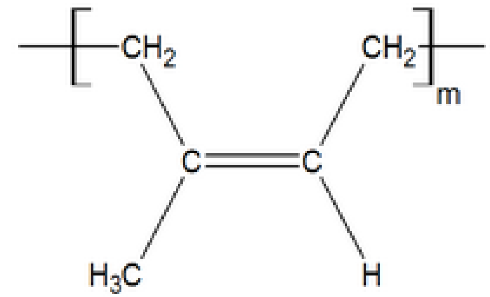


Substituent, R	Monomer	Elastomer
H	1,3-Butadiene	Polybutadiene
Cl	2-Chloro-1,3-butadiene	Polychloroprene
CH_3	2-Methyl-1,3-butadiene	Polyisoprene

Diene Elastomers



Trans isoprene



Cis isoprene

- Trans isoprene structure can
leads to relatively straight chain structure.

- can have crystalline structure (compared to cis-form)

- less elastomer property

- cis isoprene structure leads to amorphos form

- this can show little elastomer property.

Nondiene Elastomers

- A number of important elastomers do not have the unsaturated chain structure characteristic of the diene elastomer.

- These nondiene elastomers include

polyisobutylene (butyl rubber),

polysiloxane (silicone rubber),

fluoroelastomers such as Viton,

polyurethane elastomers such as Spandex, and

elastomers derived from ethylene and propylene (EP and EPDM elastomers).

Plastics

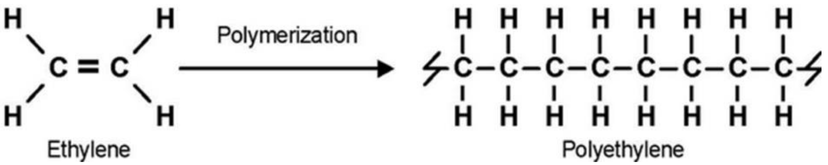
Plastics: These are polymers, which can be moulded into desired shapes by the application of heat and pressure. For example, PE, Plexiglass, PVC, PC, Teflon, etc.

Thermoplastic polymers on application of heat and pressure, initially become soft, flexible and undergo deformation.

On further heating above their melting point (Tm) they melt and flow. Such a property is called plastic deformation.

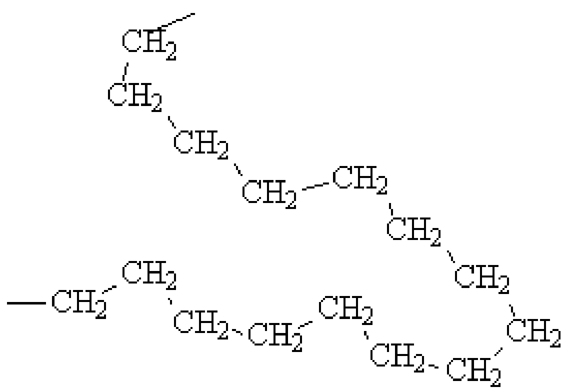
Thermoplastics exhibit plastic deformation because they are linear, closely packed and the individual chains are held by secondary forces such as van der Waals force (dispersion), and dipole- dipole interactions.

High density and low density polyethylene



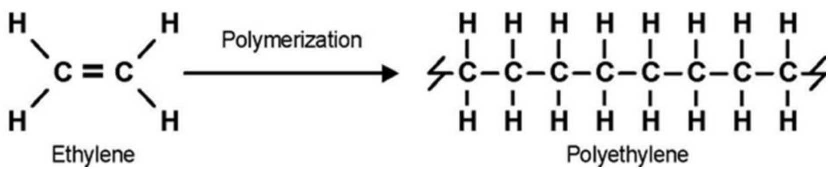
HDPE (High density polyethylene):

- The polymer chains in HDPE are more linear.
- They pack closer together, resulting in greater intermolecular forces and a more "crystalline" structure.
- HDPE has greater tensile strength than LDPE.
- Density **0.944 – 0.965 g/cm^3**



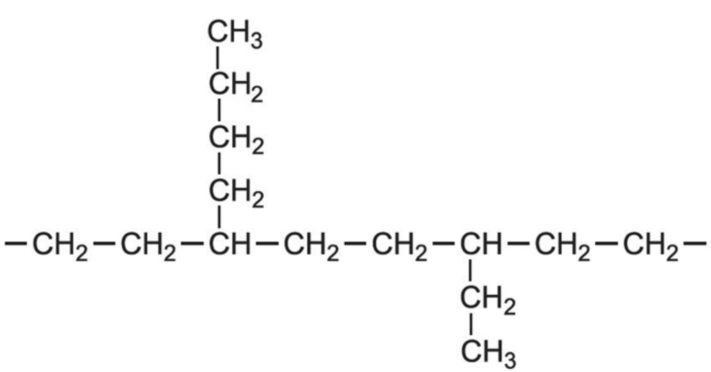
High Density Polyethylene (HDPE)

High density and low density polyethylene



LDPE (Low density polyethylene):

- The polymer chains of LDPE are highly branched
- This branching prevents the chains from stacking neatly beside each other, reducing the intermolecular forces of attraction.
- This results in a plastic that is softer and more flexible, but which also has lower tensile strength.
- Density **0.917 – 0.930 g/cm^3**

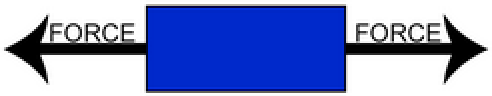


High density and low density polyethylene

Tensile strength

Tensile strength is the ability of a material to withstand a pulling (tensile) force.

It is customarily measured in units of force per cross-sectional area.



The ability to resist breaking under tensile stress is one of the most important and widely measured properties of materials used in structural applications.

Tensile strength is important in the use of brittle materials more than ductile materials.

Polyvinyl chloride (PVC)

- Synthesis:

$$n\mathrm{CH_2=CHCl} \xrightarrow{\text{Polymerize}} \underset{\text{PVC}}{-\!\!\left(\mathrm{CH_2CHCl}\right)\!\!-_n}$$

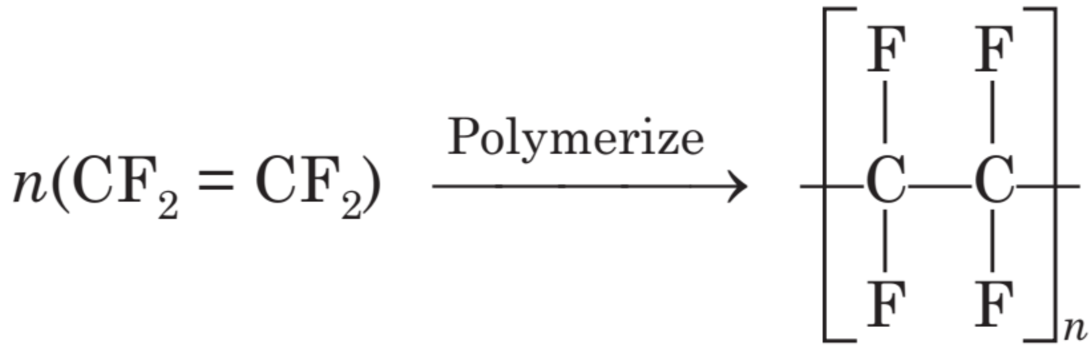
Properties: PVC is a colorless, odorless, non-flammable, chemically-inert powder. It contains 53-55% Cl_2 and softens at around 80°C.

It is resistant to water, light, O_2, inorganic acids and alkalies, oil, petrol etc., but soluble in hot chlorinated hydrocarbons.

Uses. It is the most widely used plastic. It has high rigidity and chemical resistance but brittle, so, its use is mainly in making cables, water hoses, toys, rain coats, rexin, pipes of petroleum industry, floor covering, refrigerator components, tyres, cycles and motor cycle mudguards etc.

Polytetrafluoroethene (PTFE)

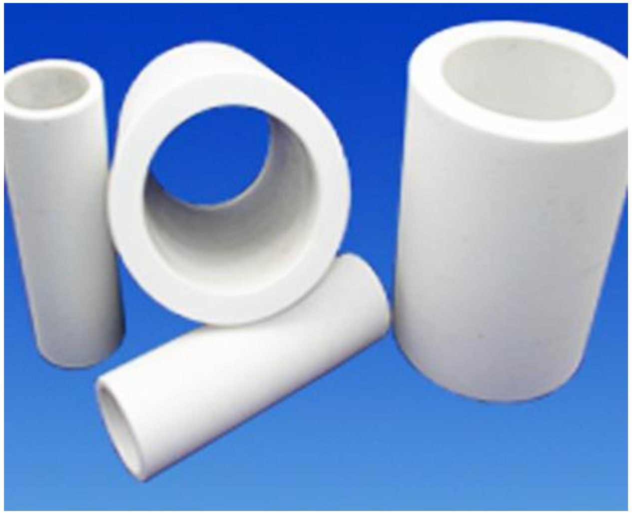
- Synthesis



Properties. Due to the presence of highly electronegative fluorine in the regular polymer structure of TEFLON strong interchain forces are present which give the material extraordinary properties like extreme toughness, high softening point (350°C), high chemical resistance, low coefficient of friction and waxy touch, good mechanical and electrical properties. Due to all these qualities, the polymeric material can be machined. It softens at about 350°C, hence at this high temperature it can be moulded applying high pressure.

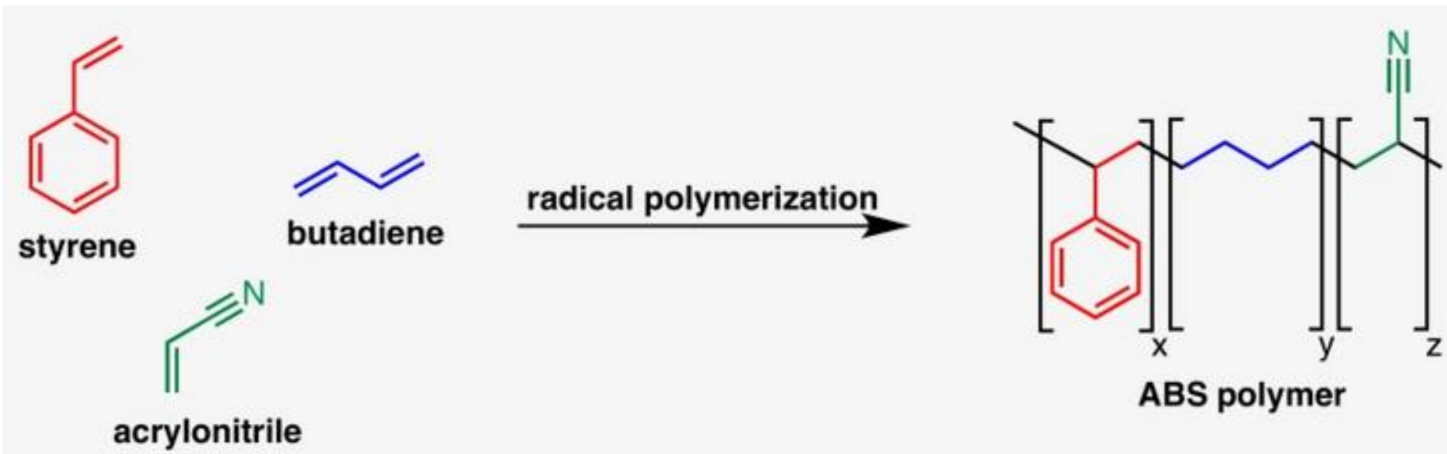
Properties and Uses of PTFE

Uses. It can be used for insulating motor, transformers, cables, wires etc. Non-stick cookware coatings are made of TEFLON. It can also be used for making gaskets, pump parts, tank linings, pipes and tubes for chemical industry, non-lubricating bearings and to make non-reactive coating.



Acrylonitrile-Butadiene-Styrene (ABS)

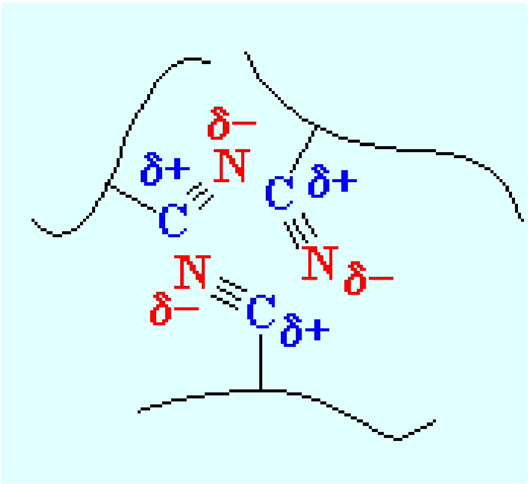
It is Graft polymer made by dissolving polybutadiene in liquid acrylonitrile and styrene monomers and then polymerizing the monomers by the introduction of free-radical initiators. It is a giant molecule predominantly made up of chains of polybutadiene backbone with styrene-acrylonitrile copolymer (SAN) branches.



ABS is a tough, **heat-resistant thermoplastic**.
Polystyrene provides ease of processing, glossiness, and rigidity.
Acrylonitrile adds chemical resistance and hardness,
and **butadiene** provides impact resistance.

ABS is the standard material of choice for the ***strong plastic cases of computers, televisions, and other household electronic goods and sometimes is used as an alternative to PVC for pipes.***

Acrylonitrile-Butadiene-Styrene (ABS)



ABS is a stronger plastic than polystyrene because of the nitrile groups of its acrylonitrile units.

The nitrile groups are very polar, so they are attracted to each other. This allows opposite charges on the nitrile groups to stabilize each other like you see in the picture on the left.

This strong attraction holds ABS chains together tightly, making the material stronger. Also the rubbery polybutadiene makes ABS tougher than polystyrene.

Fibers

These are long, thin and thread-like polymers, whose length is at least 100 times their diameter.
They do not undergo stretching and deformation like elastomers, and are linked to each other by hydrogen bonding.

In fibers, the chain mobility is reduced by very close packing of the polymer chain backbone without cross-linking.

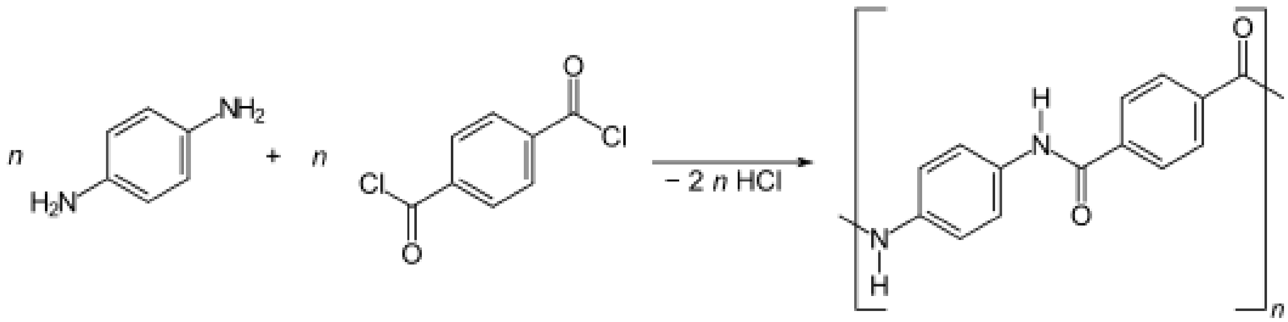
Polar groups and aromatic and cyclic rings in the backbone chain are known to impart a high strength to the polymer fiber.

For example, natural fibers such as jute, wood, silk, etc. and synthetic fibers such as nylon-6,6 and terylene.

Fibers - Kevlar Fiber

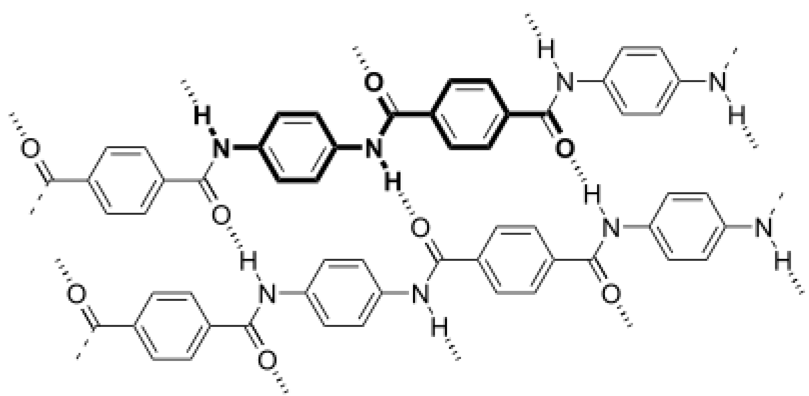
Kevlar is a heat-resistant and strong synthetic fiber.

Synthesis

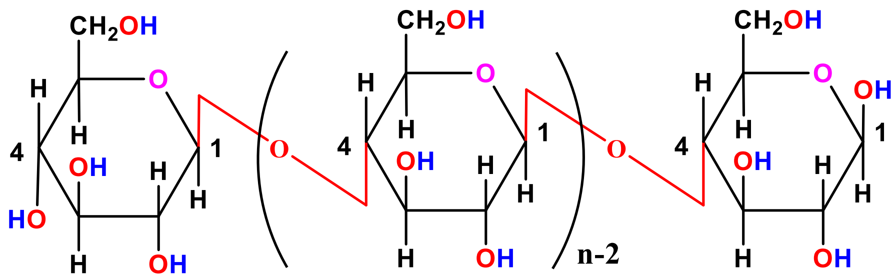


They do not undergo stretching and deformation like elastomers, and are linked to each other by hydrogen bonding.

The fiber is used to make bulletproof vests.



Fibers - Cellulose Fiber



Cellulose- a natural polymer

The monomer repeat unit present in the cellulose is β-D-Glucose

Intermolecular hydrogen bonding is present in cellulose, *therefore it has fibrous character*

The β *(1 → 4) -glycosidic linkages enables* *the cellulose polymer to form* ***intermolecular hydrogen bonding*** *with the* ***adjacent cellulose polymer units***. Large numbers of cellulose polymers blend together via intermolecular hydrogen bonding to form ***fibrous structures***. Despite cellulose is usually classified as a crystalline polymer, it has alternating crystalline and amorphous regions. The high degree of intermolecular hydrogen bonding and fibrous structures makes cellulose an insoluble molecule in water and common organic solvents.